1. Clinical Question & Core Hypothesis

Clinical Question

From the Ontologist synthesis, NMIBC patients with similar local immune infiltration (high Immunoscore/TILs) show heterogeneous benefit from intravesical BCG and emerging immunotherapies — why does immune presence sometimes fail to translate into effective anti-tumor therapy?

Working Hypothesis

In NMIBC treated with intravesical BCG, the effectiveness of anti-tumor immunity is governed by an interaction between local immune contexture (high TIL density), tumor antigenicity shaped by MSH6/MMR status (driving bTMB and neoantigen load), and the tumor-immune microenvironment's checkpoint landscape. Specifically, high TIL density will translate into productive anti-tumor responses to BCG only if the tumor presents a sufficient neoantigen load conferred by MSH6 deficiency (high bTMB/neoantigen burden). Thus: (1) TIL-high, MSH6-deficient and/or high-bTMB tumors will show superior therapy_response (lower recurrence/progression) to BCG compared with TIL-high, MSH6-proficient and/or low-bTMB tumors; (2) TIL-high tumors with intact MSH6 but low bTMB will exhibit limited benefit due to paucity of neoantigenic targets; (3) TIL-low tumors will display poor response regardless of MSH6 status or bTMB unless immune contexture is modulated; (4) checkpoint/immune landscape factors (e.g., PD-L1) and tumor-intrinsic modulators (TP53, ESR1) will modulate the magnitude of the TIL–antigenicity interaction, potentially identifying subgroups that benefit from combination therapies; (5) systemic modifiers (HDL-C, IGF-1) will influence the robustness of the local immune response and modify predicted responder status. These relationships will yield a testable interaction model: therapy_response to BCG is most favorable in TIL-high tumors with MSH6 deficiency/high bTMB, intermediate in TIL-high with intact MSH6/low bTMB, and poorest in TIL-low tumors, with PD-L1 and systemic metabolic signals shifting the thresholds for response.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	MSH6, bTMB
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

MSH6: MSH6→MMR-driven mutational load; together these provide a mechanistic chain connecting TIL presence, antigenic stimulus, and therapy-induced immune efficacy.

bTMB: bTMB→neoantigenicity/immunogenicity axis, and MSH6→MMR-driven mutational load; together these provide a mechanistic chain connecting TIL presence, antigenic stimulus, and therapy-induced immune efficacy.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	PD-L1	Supporting modifier from knowledge graph
Secondary	TP53	Supporting modifier from knowledge graph
Secondary	ESR1	Supporting modifier from knowledge graph
Secondary	HDL-C	Supporting modifier from knowledge graph
Secondary	IGF1	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

Prediction 2: In functional experiments, restoring or enhancing the anchor gene's signaling will improve immune-mediated tumor control in imaging-feature-positive models.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- PD-L1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- TP53: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.